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Paradise Lost? Estimating the Cost of Hawaii's Pandemic Border Closure with Proximal Inference

Jared Greathouse[†]

[†]*Department of Policy and Management, Georgia State University*
E-mail: `jgreathouse3@student.gsu.edu`

Summary Hawaii's March 2020 mandatory traveler quarantine sealed the state's borders at the same instant the COVID-19 pandemic collapsed global travel demand. For a single treated unit these two forces are perfectly collinear in time, so the conventional synthetic-control toolkit, which matches the treated series to a counterfactual that is itself battered by the pandemic, cannot separate the border-closure *policy* from the coincident *pandemic*. I address this with proximal / negative-control inference in two specifications, Single Proxy Synthetic Control (SPSC) and base Proximal Inference (PI), that *instrument* the latent pandemic factor using insulated employment sectors that carry the pandemic's macro-recession shock yet are immune to the border policy. The two specifications agree, recovering a large, robust tourism-demand collapse (roughly 60–75 percentage points of year-over-year growth for the visitor-volume series); but because the insulated sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly; the estimates from insulated donors alone are therefore best read as *upper bounds in magnitude* on the policy effect and are reported as a robustness band rather than as points. Recovering an actual point requires spanning the missing travel-demand factor from outside the treated economy: adding external travel-demand donors, inbound air travel to Florida and other exposed-but-open leisure destinations that were not locked down, pulls the Visitor Days estimate off its upper bound, and a convergent ensemble of such donors brackets it in a policy-centered band, taking the step from a bound to a band. The paper shows how negative-control inference yields credible causal estimates when a policy and a global shock arrive together.

Keywords: *Proximal Causal Inference, Negative Controls, Synthetic Control, Policy Evaluation, COVID-19*

1. INTRODUCTION

In 2019, tourism moved $17.75 billion in visitor spending through Hawaii's economy, underwrote $2.07 billion in state tax revenue, and sustained about 216,000 jobs, close to a third of the state's workforce Hawai'i Tourism Authority (2019). In March 2020, Hawaii deliberately switched much of that off. Among all U.S. states it enacted perhaps the most extreme and comprehensive economic shutdown of the COVID-19 pandemic (Yan et al., 2022). Where most regions contracted through voluntary distancing, fear, or uneven mandates, Hawaii chose one of the few deliberate, coordinated state-level shutdowns in the country, a policy aimed not merely at slowing the virus but at suppressing travel and commerce outright. On March 26, 2020, it imposed a mandatory 14-day quarantine on all arrivals, effectively sealing its borders. The outcome was among the starkest in the country: Hawaii went on to record the largest job losses of any U.S. state even as it held among the lowest COVID-19 case rates Bond-Smith and Fuleky (2022). This paper asks what that decision cost.

That question is not merely historical. A governor weighing whether to seal a jurisdiction, in the next pandemic or in a climate or geopolitical shock that likewise pits containment against commerce, faces a concrete tradeoff with little credible evidence on its economic side Newman and Noy (2023): the health effects of COVID-19 interventions have been studied extensively, their economic toll, especially under maximal containment, far less so. Hawaii is the clearest U.S. instance of that decision in its starkest form, a whole economy switched off to keep a virus out, and a credible price for it is what a policymaker needs to weigh the closure against the lives it aimed to protect.

Estimating that cost, however, runs into an identification problem that is easy to overlook and fatal if ignored. Hawaii closed its border in the same month the pandemic collapsed global travel demand, and neither force varied before March 2020, so for a single treated unit the *policy* and the *pandemic* are perfectly collinear in time: every post-treatment month is at once a quarantine month and a pandemic month, and no function of the observed data separates a response to one from a response to the other. This disables the two designs one would reach for first. Difference-in-differences would net out the pandemic by subtracting untreated states, but that assumes the shock lands equally everywhere de Chaisemartin and D'Haultfœuille (2022); Roth et al. (2023), and Hawaii, the most tourism-dependent state, was struck far harder than any control. Synthetic control weakens that to a factor model and reweights donors to Hawaii's pre-2020 path, but the travel-demand collapse is dormant until March 2020 and then switches on with the policy, leaving no pre-period variation to fit: even with tourism series for all forty-nine other states, the synthetic Hawaii is a no-2020 world, not a no-quarantine-but-still-pandemic one, and differencing charges the entire post-March collapse to the border closure. Section~3.1 makes this formal; the failure is a property of the coincident shock, not a shortage of data.

This paper takes that identification problem seriously and resolves it with the right tool. The pandemic confounder is a time-varying latent factor that moves the outcome and coincides with treatment; it cannot be conditioned away because it is unobserved, and it cannot be matched away because it moves any donor too. The *proximal* answer Miao et al. (2018); Tchetgen Tchetgen et al. (2024) is to instrument it: find observable negative controls that are driven by the same pandemic factor but carry no causal path from the treatment, and use them to subtract the pandemic out of the counterfactual. Here those negative controls are insulated employment sectors, hit by the pandemic recession but not downstream of whether tourists can reach Hawaii. I fit two proximal specifications, Single Proxy Synthetic Control (SPSC) Park and Tchetgen Tchetgen (2025) and base Proximal Inference (PI), and report both, so the estimate does not hinge on one estimator's machinery.

Empirically, the two specifications agree: fit on the insulated proxies, SPSC and PI recover the same large tourism-demand collapse, so the result rests on the identification rather than on one estimator. This delivers the first credible, if bounded, price for a maximal pandemic border closure, a collapse on the order of 60 to 75 percentage points of year-over-year growth for the visitor-volume series. What they identify is a bound, and I treat it as one. The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor, so the tourism estimates absorb both the policy and the demand loss and bound the border-closure effect from above. I report them as a specification band and check each against a relevance diagnostic. Converting that bound into a point requires a proxy that carries the missing travel-demand factor, and I supply one from outside the treated economy: inbound air travel to open, no-lockdown destinations, Florida foremost, which pulls the Visitor Days estimate off its upper bound to a policy-centered point.

The remainder of the paper proceeds as follows. Section 2 reviews prior research on early pandemic interventions and the methodological challenge of studying one-off, jurisdiction-wide policies. Section 3 states the coincident-shock identification problem and the two proximal estimators. Section 4 describes the data. Section 5 reports the proximal estimates and their identification diagnostics, the robustness band, and a set of falsification checks. Section 6 discusses implications for public policy and crisis response.

2. BACKGROUND

2.1. Tourism in Hawaii

Hawaii's economy transitioned from a plantation-dominated model in the late 19th and early 20th centuries to tourism as the primary driver after statehood. The sugar and pineapple industries, controlled by a few powerful firms known as the "Big Five," peaked in the 1960s but declined due to global competition, labor costs, and diversification efforts. Statehood in 1959, combined with the introduction of jet air travel, spurred a tourism boom, with visitor numbers surging from 171,000 in 1958 to over 6.7 million by 1990 Croix (2021). This shift modernized the economy, increased population growth, and generated substantial revenue, but it also created dependency on external factors like air travel and international markets.

The scale of that dependence is worth making concrete. On any given day in 2019, Hawaii hosted about 249,000 visitors spending \$48.6 million, with O'ahu (\$22.4 million daily) and Maui (\$14.0 million) bearing the brunt Hawai'i Tourism Authority (2019); leisure and hospitality employment, a proxy for tourism-related jobs, held around 190,000–200,000 pre-2020. This reliance amplifies the state's exposure to shocks even as it funds public services and infrastructure.

Hawaii implemented some of the earliest and most aggressive COVID-19 containment policies in the United States. On March 4, 2020, Governor David Ige issued an emergency proclamation officially recognizing the virus as a statewide threat (Maui Now Staff, 2020). Within weeks, the state began taking unprecedented measures alongside states like California (Friedson et al., 2021). On March 17, Governor Ige publicly urged all visitors to suspend their trips to Hawaii for 30 days, anticipating substantial economic damage to the state's tourism-dependent economy. *"We do know there will be significant impact,"* he stated, though no concrete estimates were available at the time (Nakaso, 2020). A sweeping stay-at-home order followed on March 25, banning non-essential activities, closing non-critical businesses, and effectively shutting down day-to-day life and travel across the islands (HNN Staff, 2020). These actions were implemented alongside a mandatory 14-day quarantine on all arrivals, passed the same week, which further intensified the shutdown and isolated the state from both domestic and international travel. Despite the unprecedented scope of these interventions, no formal quantitative analysis has yet estimated the short-run economic effects of Hawaii's approach. This paper provides such an assessment.

2.2. Literature Review

A large early literature makes the public-health case for acting fast, and the lesson is consistent: early, aggressive containment slows transmission and buys hospital capacity. Friedson et al. (2021) find that California's statewide shelter-in-place order, the first in the country, averted 160 to 195 COVID-19 cases per 100,000 and as many as 1,566 deaths within a month, at a back-of-the-envelope cost of roughly 650 to 700 jobs per life saved. Yang (2021) estimates that Wuhan's 76-day cordon cut inbound mobility by about 60% and halved the growth of new cases within days, and Cerqueti et al. (2021), using an augmented synthetic control, reach the same sooner-the-better verdict for Italy's national lockdown. Timing is what pays: Bayat et al. (2020) estimate that locking down ten days earlier would have cut New York deaths by as much as 80%, and Sobiech Pellegrini (2022) and Li and Shankar (2023) document the mirror image, sharp case rebounds after premature reopenings in Santa Catarina and Texas. These designs price the health payoff of containment, and the California study even prices the jobs traded for it, but they do so for interventions milder than a full border seal and lean on comparison regions that a coincident global shock would contaminate.

The economic side of the ledger is thinner, and where it exists it tends to find the damage real but transitory, and mostly not the policy's own doing. Ke and Hsiao (2022), with a panel-data counterfactual, estimate that Hubei's 76-day lockdown cut first-quarter 2020 GDP by about 37%, though the economy rebounded within months of reopening, and Gong et al. (2024) put the output cost of China's stricter 2022 zero-COVID phase at 3.9% of GDP. The recurring result is that the observed economic hit is dominated by the common pandemic shock rather than the local policy. Lee and Yang (2022) find the pandemic erased 1.1 million Korean jobs by April 2020, but only 9% of that national loss traces to regional COVID intensity, the remaining nine-tenths to a nationwide common effect; Gibson and Sun (2023) likewise attribute only 32% of the U.S. spike in new jobless claims to stay-at-home orders, the rest to the general spread of the virus and collapsing consumer confidence; and Osman et al. (2022) can identify state-level effects at all only by leaning on the cross-state policy variation that a uniform national shock erases. That decomposition is the crux of the present paper: when the observed collapse is mostly the coincident shock, a credible policy estimate must separate the two, because differencing against a control the same shock has already moved cannot. Set against these costs, containment also carried some economic and welfare upside, from the pollution-mortality co-benefits of Wuhan's lockdown Cole et al. (2020) to the GDP losses that Europe's vaccine-certificate mandates averted Oliu-Barton et al. (2022).

For all its breadth, this literature leaves the decision at its extreme end essentially unpriced. The question a policymaker faces is not whether containment works but what its most aggressive form costs, and that figure is what the record lacks. The gap has teeth because the politics cut against early action: as Cerqueti et al. (2021) observe, a containment that succeeds looks excessive in hindsight, so the officials who act first are precisely those most in need of a credible after-the-fact accounting to defend the call. Hawaii is the clearest U.S. instance of that most-aggressive option, and this paper supplies the missing price for a "zero-COVID" style border (Yuan, 2022).

While much has been learned about the effects of pandemic interventions, measuring their economic impact remains hard, and Hawaii pushes that difficulty to an extreme.

Most evaluations rely on difference-in-differences or synthetic control, and both lean on cross-sectional variation in policy, some counties or states acting while others do not, or acting later, to supply the untreated or not-yet-treated comparison the design needs (e.g., Yang, 2021; Gong et al., 2024; Bilgel, 2022). Callaway and Li (2023) show that even this variation is treacherous during an epidemic, because the timing-based identifying assumptions are hard to reconcile with the dynamics of contagion. And it vanishes entirely for a single treated unit hit by a global shock: every other U.S. state and comparable island economy faced its own pandemic disruption, so there is no untreated Hawaii to borrow across space. Ke and Hsiao (2023) confront a version of the same coincident-shock problem, developing a panel method for data subject to multiple treatment effects that disentangles a global pandemic's impact from a specific, coincident one, and Li et al. (2023) apply the same panel-plus-time-series logic to separate a Shanghai housing policy from the pandemic. The only prior study of Hawaii's pandemic economy, Bond-Smith and Fuleky (2022), documents the depth of the decline but compares it against a pre-pandemic forecast rather than a policy counterfactual.

One response to the missing-donor problem is to borrow strength across *time* rather than space: the Synthetic Historical Control (SHC) method of Chen et al. (2024) reconstructs the treated unit's counterfactual from its own pre-treatment trajectory. But borrowing across time does not, by itself, solve the deeper problem here. Whether the counterfactual is built from other states (classical SCM) or from Hawaii's own history (SHC), it is a world with *neither* the quarantine *nor* the pandemic; extrapolating it into 2020 and differencing against the observed series attributes the entire post-March collapse, pandemic and policy together, to the treatment. The issue is the *coincidence* of two shocks in a single unit, whatever the source of the donor information, so I do not use own-history extrapolation as the estimator and turn instead to the proximal / negative-control approach.

That is precisely the structure the *proximal / negative-control* literature was built to handle. Negative controls, variables driven by an unmeasured confounder but shielded from the treatment, have long been used to detect confounding Lipsitch et al. (2010); proximal causal inference turns them into an identification strategy, using proxies of a latent confounder to recover causal effects that ordinary adjustment cannot Miao et al. (2018); Cui et al. (2024); Tchetgen Tchetgen et al. (2024); Shi et al. (2020). Park and Tchetgen Tchetgen (2025) adapt this machinery to the synthetic-control setting as Single Proxy Synthetic Control, and Liu et al. (2024) develop a related surrogate-based proximal estimator. I apply proximal inference, SPSC and base PI, to Hawaii because it is the one approach in the synthetic-control family that concedes the pandemic confounder is present and unobservable and instruments it, rather than matching or conditioning it away.

3. METHODOLOGY

3.1. The identification problem: a coincident common shock

We know how Hawaii's economy looked before the border closure, and what it did once it mandated the quarantine; but estimating the policy's cost requires a counterfactual, an estimate of how Hawaii's tourism sector would have evolved absent the closure. To build one, we adopt the comparative case study framework, comparing a single treated unit against a set of control units, or donors, that did not enact the policy. Index the N units by j, with $j=1$ the treated unit (Hawaii) and the other N_0 the control units $\mathcal{N}_0$, and index the T months by t, collected in $\mathcal{T}=\{1,\ldots,T\}$ and split at the final pre-quarantine month T_0 into $\mathcal{T}_1=\{t\le T_0\}$ and $\mathcal{T}_2=\{t>T_0\}$; y_{jt} is unit j's outcome in month t. Write y^1_{1t} and y^0_{1t} for Hawaii's month-t outcome with and without the closure. Only one is ever seen: with $d_t=\mathbf{1}\{t>T_0\}$ the treatment indicator, $y_{1t}=d_t\,y^1_{1t}+(1-d_t)\,y^0_{1t}$.

Synthetic control builds the counterfactual as a weighted average of the donors. Stacking the treated unit's path into $\mathbf{y}_1\in\mathbb{R}^T$ and the donors' paths columnwise into $\mathbf{Y}_0\in\mathbb{R}^{T\times N_0}$, it returns the weight vector $\mathbf{w}$ that minimizes the pre-treatment mean squared error between the treated unit and the weighted donors,

$$\mathbf{w}^*=\underset{\mathbf{w}\in\Delta^{N_0}}{\operatorname{argmin}}\ \big\|\mathbf{y}_1-\mathbf{Y}_0\mathbf{w}\big\|_2^2. \tag{3.1}$$

The weights are restricted to the probability simplex $\Delta^{N_0}=\{\mathbf{w}\in\mathbb{R}^{N_0}_{\ge 0}:\sum_{j\in\mathcal{N}_0}w_j=1\}$. Carried past T_0, the fitted values $\mathbf{Y}_0\mathbf{w}^*$ of (3.1) are the counterfactual for the treated unit, and the quarantine's cost is the average treatment effect on the treated over the post-intervention period,

$$\text{ATT}:=\frac{1}{n}\sum_{t\in\mathcal{T}_2}\big(y^1_{1t}-y^0_{1t}\big),\qquad n=|\mathcal{T}_2|. \tag{3.2}$$

Whether that estimate is credible depends on the data-generating process. The usual justification for synthetic control is the interactive fixed-effects model Bai (2009); Abadie et al. (2010),

$$y^0_{1t}=\boldsymbol{\mu}_1^\top\boldsymbol{\lambda}_t+e_{1t},\qquad y_{jt}=\boldsymbol{\mu}_j^\top\boldsymbol{\lambda}_t+e_{jt},\quad j\in\mathcal{N}_0, \tag{3.3}$$

where $\boldsymbol{\lambda}_t\in\mathbb{R}^r$ collects time-varying factors common to all units, $\boldsymbol{\mu}_j$ collects unit-specific, time-invariant loadings, and $\mathbb{E}[e_{jt}\mid\boldsymbol{\lambda}_t]=0$ is idiosyncratic error. Under this model, synthetic control is the standard tool, and the theory is precise about when it works. Differencing the treated series against any weighted counterfactual leaves, after some algebra,

$$y_{1t}-\sum_{j\in\mathcal{N}_0}w_jy_{jt}=\underbrace{\big(y^1_{1t}-y^0_{1t}\big)}_{\text{policy effect}}+\underbrace{\Big(\boldsymbol{\mu}_1-\textstyle\sum_{j\in\mathcal{N}_0}w_j\boldsymbol{\mu}_j\Big)^\top\boldsymbol{\lambda}_t}_{\text{factor-mismatch bias}}+\big(e_{1t}-\textstyle\sum_{j\in\mathcal{N}_0}w_je_{jt}\big), \tag{3.4}$$

the policy effect plus a factor-mismatch bias, the residual loading gap $\boldsymbol{\mu}_1-\sum_j w_j\boldsymbol{\mu}_j$ projected onto the common factors, plus idiosyncratic noise Abadie et al. (2010); Ferman and Pinto (2021); Zhang et al. (2022); Hollingsworth and Wing (2022). The guarantee behind synthetic control is that a long pre-period with a close pre-treatment fit drives that loading gap to zero, forcing the weights to reproduce Hawaii's loadings,

$$\boldsymbol{\mu}_1=\sum_{j\in\mathcal{N}_0}w_j\,\boldsymbol{\mu}_j. \tag{3.5}$$

For that guarantee to hold however, the same factors and loadings must govern the pre- and post-treatment periods, so that a fit achieved before treatment still holds after it. In the Hawaii case the condition fails. In March 2020 a new coordinate of $\boldsymbol{\lambda}_t$ switches on, the effect of the pandemic, absent from the data before the quarantine and present after, and so perfectly collinear with the treatment over the post-period. The weights are fit on the pre-period, where this factor is silent, so they cannot be chosen to reproduce Hawaii's loading on it: the factor-mismatch term in (3.4) no longer vanishes, and because the factor is unobserved and absent in sample, the bias cannot even be estimated. Practically, this means that even if we had noiseless, observed tourism data across the remaining United States, a good pre-treatment fit built from the donor states no longer certifies the counterfactual, and no reweighting of pre-period donors can subtract a shock the pre-period never saw.

3.2. Proximal inference: single-proxy and base specifications

Fortunately, one solution to this collinearity is the relatively new strand of synthetic control that employs proximal inference Miao et al. (2018); Tchetgen Tchetgen et al. (2024); Shi et al. (2026). The idea is to find a negative control, or proxy: a variable we can use to net the pandemic out of the counterfactual and leave the policy's effect behind. For a proxy to do that, it must be both relevant and exclusionary. A relevant proxy has a strong relationship with the outcome of interest over the pre-policy period, so that it carries a genuine reading of the confounding pandemic factor. An exclusionary proxy is one the policy does not affect, so that reading stays uncontaminated by the treatment we are trying to measure. A variable with both properties moves with the pandemic but not with the border closure, which is exactly what lets the pandemic be subtracted from the counterfactual while the policy effect stays in, pulling apart two forces the collinearity had fused into one.

To state these conditions precisely, collect the candidate proxies at month t in a vector $\mathbf{p}_t$. Relevance asks that some combination of the proxies track the treated unit's untreated outcome,

$$\operatorname{Cov}\big(h^\star(\mathbf{p}_t),\, y_{1t}^0\big) \neq 0 \quad \text{for some } h^\star : \mathbb{R}^{N_0} \to \mathbb{R}, \qquad t \in \mathcal{T}_1, \tag{3.6}$$

the proxy analogue of a strong first stage, whose empirical footprint is the pre-treatment fit; a proxy that does not move with y_{1t}^0 carries no reading of the factor and cannot help. Relevance and exclusion together secure a clean reading of the pandemic, but a reading is not yet a counterfactual. What connects the two is a bridge function: a function $h^\star$ of the proxies that, in expectation, equals the untreated outcome,

$$y_{1t}^0 = \mathbb{E}\big[\, h^\star(\mathbf{p}_t) \,\big|\, y_{1t}^0 \,\big] \quad \text{almost surely}, \qquad t \in \mathcal{T}, \tag{3.7}$$

so that passing the proxies through $h^\star$ returns what Hawaii's tourism would have done absent the closure. In short, a good proxy is one the pandemic drives strongly (relevance) and the policy leaves alone (exclusion), tied to the treated counterfactual through a bridge function.

Two estimators turn the proxies into a counterfactual, differing only in how they pin down the bridge weights $\mathbf{w}$ in $h^\star(\mathbf{p}_t) = \mathbf{p}_t^\top \mathbf{w}$. Single Proxy Synthetic Control Park and Tchetgen Tchetgen (2025) uses the treated unit's own pre-treatment history as the instrument and fits the bridge by ridge-regularized GMM on the de-trended pre-period series,

$$\big(\widehat{\boldsymbol{\eta}}, \widehat{\mathbf{w}}_\varrho\big) = \underset{\boldsymbol{\eta},\mathbf{w}}{\operatorname{argmin}} \ \Big\{ \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta},\mathbf{w})^\top \,\widehat{\boldsymbol{\Omega}}\, \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta},\mathbf{w}) + \varrho\, \|\mathbf{w}\|_2^2 \Big\}, \tag{3.8}$$

with $\overline{\boldsymbol{\Psi}}$ the averaged moment conditions, $\widehat{\boldsymbol{\Omega}}$ a weight matrix, and the ridge penalty ϱ (cross-validated) stabilizing the weights when the single instrument leaves them under-identified. Base Proximal Inference (PI) is the donors-only estimator of Shi et al. (2026), which in this implementation reduces to a least-squares fit of Hawaii's pre-period series on the donors with a GMM/HAC standard error. Either way the fitted bridge gives the counterfactual $\widehat{y}^0_{1t} = \mathbf{p}_t^\top \widehat{\mathbf{w}}$ and the ATT (3.2) as the average post-period gap $y_{1t} - \widehat{y}^0_{1t}$; the estimating equations, penalty selection, standard errors, and conformal band are given in Park and Tchetgen Tchetgen (2025) and implemented in mlsynth. Because the treated unit supplies a single instrument against the donors, the analytic standard errors are fragile here, so I lean on the specification band of Section~5 for uncertainty; sharing the identification but not the algorithm, the two estimators' agreement there reflects the proxy design rather than one estimator's machinery.

The proxies fall into two groups, both detailed in Section~4. The first is the $N_0 = 5$ insulated employment sectors: they fell in the pandemic recession, so they carry its macro-recession factor (relevance), but they do not depend on whether visitors can reach Hawaii, so the border closure leaves them untouched (exclusion). The second is a set of external travel-demand donors, inbound air travel to exposed-but-open leisure destinations (Florida foremost), which carry the pandemic's travel-demand-collapse factor while lying outside Hawaii's policy. Both enter the same proxy matrix $\mathbf{P} := \mathbf{Y}_0$, with month-$t$ row $\mathbf{p}_t$; a travel donor widens $\mathbf{P}$ to span a second factor rather than adding a control of a different kind. The split governs what the estimator can recover: the insulated sectors span only the macro-recession factor, so on them alone every tourism estimate is an upper bound in magnitude on the policy effect, absorbing the travel-demand collapse, and adding a travel-demand donor is what turns that bound into a point (Section~5.3). I summarize each fit by its pre-treatment root-mean-squared error $\mathrm{RMSE}_{\mathrm{pre}} = \{T_0^{-1} \sum_{t \in \mathcal{T}_1} (y_{1t} - \widehat{y}^0_{1t})^2\}^{1/2}$, small relative to the effect when the reconstruction is informative rather than an artifact.

4. DATA

The analysis joins three public sources on calendar month and works throughout in year-over-year (YoY) percentage growth. The YoY transformation removes the strong seasonality of Hawaii tourism and employment, puts series measured in different units on a common scale, and lets each treatment effect be read directly as a percentage-point change in growth. Every series is stored as a raw monthly level and differenced to YoY growth at estimation time, so the panel retains levels for any later recovery analysis rather than shipping pre-differenced series. Hawaii's Department of Business, Economic Development & Tourism (DBEDT) supplies the treated tourism series and the insulated employment sectors; the U.S. Bureau of Transportation Statistics (BTS) supplies the external air-travel proxies. The estimation window is described at the end of the section.

4.1. Treated outcomes

The treated series are Hawaii's five tourism outcomes from DBEDT: the visitor-volume and price measures `Visitor Arrivals`, `Visitor Days`, `Occupancy`, `Mean Daily Rate`, and `Revenue per Available Room`. Each is stored as a raw monthly level (visitor counts, occupancy percentages, and dollar rates) and differenced to year-over-year growth at estimation time. Every one is downstream of the border closure: each moves when visitors' ability to arrive changes, so each can only be a treated outcome, never a control.

4.2. Insulated-sector proxies (the macro negative controls)

The negative-control donors are five insulated employment sectors drawn from the DBEDT seasonally-adjusted Current Employment Statistics, the monthly job count by industry Hawai'i Department of Business, Economic Development and Tourism (2025), expressed in YoY growth: `NatRes_Constr_Emp` (mining, logging and construction), `Wholesale_Emp` (wholesale trade), `Financial_Emp` (financial activities), `HealthCare_Emp` (health care and social assistance), and `Government_Emp`. These industries were hit by the pandemic's macro-recession shock, falling roughly 7–12% in the 2020 trough, but do not depend on whether tourists can physically enter Hawaii, so no causal path runs from the quarantine to them. They are the empirical negative controls of Section~3.2.[1] Figure 1 contrasts the two groups: the treated tourism series collapse by 50–99% at the border closure while the insulated donors dip only modestly, the visual signature of a factor structure in which the donors carry the macro shock and the tourism series carry the macro shock plus the travel-demand collapse plus the policy.

4.3. Open-destination air travel (the travel-demand negative controls)

The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor (Section~3.2), so a second class of proxy is needed to carry that factor: inbound air travel to *exposed-but-open* destinations, places whose air traffic collapsed with the pandemic yet which imposed no traveler quarantine, and so register the travel-demand shock without registering Hawaii's policy. I build these from the BTS Airline On-Time Performance database Bureau of Transportation Statistics (2024), scraped from the agency's airport-statistics portal (https://www.transtats.bts.gov/airports.asp): monthly counts of arriving flights by airport, aggregated to a statewide total for Florida and to the individual airport for the Tampa (TPA), Orlando (MCO), and Phoenix (PHX) leisure metros, then converted to YoY growth. Florida is the archetype, a large, tourism-dependent state whose government kept its borders open through 2020, and the three metros diversify the proxy across independent leisure markets in states that likewise never locked down. Their identifying role, and why I fit them one at a time rather than en masse, is developed in Section~5.3.

4.4. Panel and estimation window

The treatment indicator `Mandatory Quarantine` switches to 1 in March 2020, giving 10 post-treatment months (March–December 2020); the post-window is capped at December 2020 so the extended raw panel cannot leak recovery months into the estimate. Although

[1]The five are the most insulated supersectors in the CES taxonomy Hawai'i Department of Business, Economic Development and Tourism (2025). I screened the full taxonomy for the negative-control property and deliberately *excluded* the sectors that are downstream of visitor arrivals and would therefore violate it: retail trade (visitor spending), transportation, warehousing and utilities (which bundles air transportation), arts, entertainment and recreation, accommodation and food services, and real estate, rental and leasing (car and vacation rental). Empirically these fell 23–62% in the 2020 trough, tracking the tourism collapse rather than the macro recession, which is exactly the contamination that disqualifies them as controls. A second tier of genuinely insulated sectors, manufacturing, professional and business services, and information, could enlarge the donor pool, but their pre-2020 comovement with the tourism series is no stronger than the five retained donors' ($|\text{corr}| \leq 0.4$), so they do not sharpen identification of the tourism (travel-demand) factor. The one donor that does carry that factor is external, inbound air travel to Florida, added in Section 5.3.

the panel ships raw levels from 1990 onward, the analysis truncates the pre-period to a recent window beginning January 2014 (74 pre-treatment months). Three decades of stale comovement is less relevant to the 2020 counterfactual than the recent history, and the 2014 window empirically sharpens the pre-treatment fit and yields non-degenerate GMM standard errors that the full-history-plus-detrend combination did not. The full-panel version is available by removing the truncation and enters the robustness band of Section~5.

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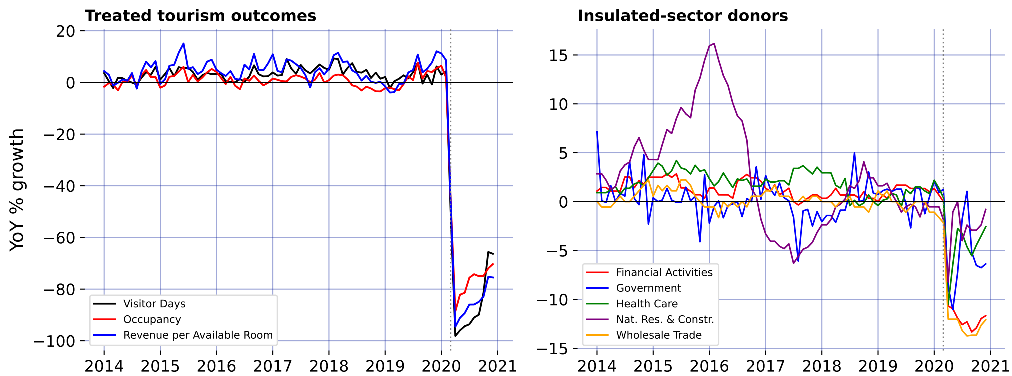

Figure 1: The negative-control premise: treated tourism outcomes collapse while the insulated-sector donors barely dip.

5. RESULTS

5.1. The pre-treatment fit: relevance

Each estimate is only as good as the bridge's pre-treatment fit, the empirical footprint of the relevance condition (3.6); Figure 3 draws the per-outcome counterfactuals over time. Relevance holds where it matters: Visitor Days, Visitor Arrivals, Occupancy, and Revenue per Available Room all reconstruct to pre-treatment RMSEs of at most 3.6 growth-rate points (for Visitor Days, 2.6 points against a $\sim$ 70-point effect), so the collapse the estimator reads off is not an artifact of a poor fit.

The GMM standard errors are usable for the visitor series on this 2014 window ($\sim$ 3–8 points), unlike the full-history window where de-trending a near-interpolating fit collapses them toward zero. Even a well-behaved SE captures only sampling noise, not the extrapolation risk that dominates here, so I lean on the specification band below.

5.2. The insulated-only estimate: a bound

Fit on the insulated-sector proxies, the two headline specifications agree: for Visitor Days, SPSC returns an ATT of -68.5 percentage points and base PI -76.9, and across the five headline tourism outcomes they never differ by more than 9 points. A regularized instrumental-variable bridge (SPSC) and a plain least-squares fit (PI) landing on the same effect from the same proxies makes it a property of the proxy design, not of one estimator. But these are *upper bounds* in magnitude on the policy effect: the insulated sectors span the macro-recession factor, not the travel-demand collapse (Section~3.2), so the tourism loading $\boldsymbol{\mu}_1$ lies outside the donor span (3.5) along that direction and the ATT still carries it. The observed drop is policy plus travel-demand loss; closing the gap to a point needs a travel-demand donor from outside the treated economy, next.

5.3. Completing the bound: an ensemble of exposed-but-open destinations

The open-destination air-travel proxies documented in Section~4, inbound flights to Florida and to the Tampa, Orlando, and Phoenix leisure metros, all exposed-but-open, are what close the travel-demand gap that leaves the insulated-only estimates as bounds. Rather than lean on one, I use the whole small ensemble; Florida is the archetype, its air traffic down about 70% at the 2020 trough. Each enters the long panel as its own donor unit (`group = travel-demand`), and, this matters, I fit them *one at a time*, the insulated five plus a single open-destination donor, not all at once (I return to why below).

An open destination satisfies the proxy conditions from the other side: its inbound air travel loads on the same travel-demand collapse the insulated sectors miss (relevance), and as a separate economy under no Hawaii quarantine it carries that factor without importing the treatment (exclusion).

The exclusion that does the most work, though, is from Florida's *own* policy, and it is what makes Florida a deliberate choice rather than a convenient one. For the proxy to read the pandemic's voluntary travel-demand collapse cleanly, Florida's air-travel decline must not itself be an artifact of a Florida lockdown, and it is not: Florida's mandates were nominal and short-lived. Bollyky et al. (2023) measure each state's policy stringency by the share of pandemic days its mandates (business and school closures, mask and stay-at-home orders, gathering limits) were in force and place Florida at the permissive end, imposing fewer restrictions on businesses and individuals than most states Liu et al. (2024); the state's response was politically driven and centralized to the point of sidelining local emergency-management authority, reopening early and later barring its own localities from reimposing restrictions Witkowski et al. (2023). Behavior matched the policy. At the March 2020 peak of the shock Florida's beaches drew spring-break crowds in the thousands even as federal guidance urged against gatherings, phone-mobility data tracked those crowds dispersing across the eastern United States Business Insider (2020), genomic surveillance later traced new-variant introductions to that spring-break travel Napolitano et al. (2024), and in Pinellas County, part of the Tampa Bay leisure market that furnishes one of the donors here, ethnographers recorded beach tourism continuing through the pandemic with its protocols widely unobserved Porter and McIlvaine-Newsad (2022). That inbound flights to Florida collapsed anyway, with little binding policy to explain it, is precisely the point: the decline is the voluntary travel-demand factor itself, which is exactly what the proxy must carry. The criterion has teeth, and it is not met by every warm-weather destination: Puerto Rico, a fellow island otherwise tempting as a donor, imposed a genuine islandwide stay-at-home order in mid-March 2020, so its tourism collapse is contaminated by its own policy and it is set aside for that reason.

The remaining objection is that a proxy like Florida still brings idiosyncratic confounders of its own, its hurricane season, its larger drive-to share of visitors. Two features contain that objection. First, the bridge weight is fit on the clean pre-period, so a Florida-specific shock that does not track Hawaii visitor volume before 2020 enters as fit noise and is down-weighted; what licenses the weight is the pre-2020 correlation of +0.35 and the pre-treatment RMSE it earns, and the post-period behavior plays no part in the fit. Second, and decisively, the point does not rest on Florida alone: four independent markets across two states enter singly and converge on the same band, so for an idiosyncratic confounder to move the estimate it would have to be common to all four and aligned with the March 2020 quarantine, which is the pandemic travel-demand collapse itself, the very factor the donors are meant to carry. Where a proxy fails relevance the procedure reports it rather than forcing a fit, as with Occupancy below, where Florida is down-weighted to zero.

Figure 2 reports the convergence, and the headline is robustness through agreement. For Visitor Days, every one of the four independent open-destination donors pulls the estimate off the insulated-only bound of -68.5 points into a policy-centered band of roughly -52 to -65 points, and each one *improves* the pre-treatment fit rather than degrading it. That four proxies chosen independently, different cities, two different states, move the estimate the same way and by a similar amount is far stronger evidence that the observed drop is policy plus a real, separable travel-demand collapse than any single proxy could be. Florida gives the cleanest single fit (its Visitor Days pre-treatment RMSE falls from 2.58 to 1.92, bridge weight $w_{\mathrm{FL}} = 0.14$), consistent with its pre-2020 correlation of +0.35 with Hawaii visitor volume.

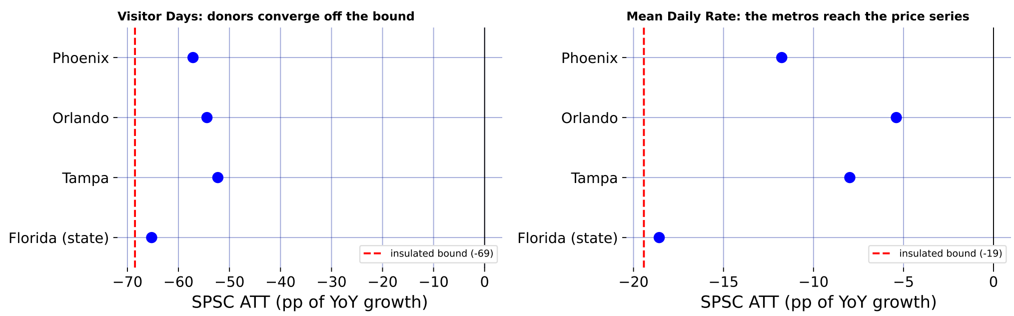

Figure 2: The open-destination ensemble: each donor, fit singly, pulls the estimate off the insulated-only bound.

The ensemble also reaches an outcome a single volume proxy could not. Florida is a flight-*volume* series, so it comoves with Hawaii *visitor volume* but not with hotel prices, and leaves Mean Daily Rate essentially at its insulated-only value of -19.4 points. Phoenix, a desert-resort market whose room rates swing with leisure demand, does comove with Hawaii's daily room rate (pre-2020 correlation +0.36), and adding it pulls Mean Daily Rate from -19.4 toward -11.8 points while *improving* the fit (pre-treatment RMSE 2.38 to 1.82). Different open destinations carry the travel-demand factor as it manifests in different outcomes, volume for the Florida metros, price for Phoenix, so a small, well-chosen ensemble reaches further into the tourism block than any one proxy.

This is why I fit the donors *singly.* Added en masse, the four near-collinear proxies induce large offsetting ridge weights that *degrade* the fit where a single well-chosen donor helped most, the price series (Revenue per Available Room's pre-treatment RMSE rises from 3.61 to 4.41). Independent proxies read one at a time corroborate as a convergence band; collapsed into one super-donor, the collinearity punishes them.

Figure 3 is the summary figure: for every headline outcome it draws the observed series, the insulated-only SPSC counterfactual (macro factor only, the upper bound, with its conformal spikes), the counterfactual after adding Florida (travel-demand factor as well), and base PI's insulated path as a dotted overlay sitting almost on the SPSC insulated line, making the two-specification agreement visible. The wedge between the two SPSC lines is where the travel-demand donor works, wide for the visitor-volume series, absent for Occupancy where Florida is correctly down-weighted to zero. No single wedge is a clean decomposition, though; each +donor fit re-solves all weights jointly, so the honest object is the convergence across donors in Figure 2.

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5.4. *Robustness and falsification*

5.4.1. A specification band, not a point Because SPSC and base PI leave the bridge weights unconstrained, dropping the simplex restriction for the flexible $\mathbb{R}^{N_0}$ of Doudchenko and Imbens (2016), the counterfactual can extrapolate outside the donor hull, and the GMM sandwich, degenerate with a single instrument against five donors on a near-interpolating pre-window, reports a sampling noise that collapses toward zero. The uncertainty that actually binds is this extrapolation, or specification, risk, so I report the headline effects as a band built from specification sensitivity: de-trending on versus off, the B-spline degrees of freedom, the polynomial basis degree, dropping each of three insulated donors individually (the two largest-weight sectors and construction, the proxy most plausibly linked to tourism), and the full-1991 versus 2014 window. Figure 4 collects the minimum, median, and maximum SPSC ATT across these eight specifications for the three best-behaved outcomes.

For the visitor-volume series the band runs roughly from -74 to -57 points (median -68), so the defensible insulated-only statement is a $\sim$ 60–75-point upper-bound collapse rather than any single figure.

5.4.2. How much would an exclusion violation have to bite? The identifying content of the insulated proxies rests on exclusion: the border policy must not move them directly (Section~3.2). That is an assumption. Rather than take it on faith I bound the consequence of its failure, a sensitivity analysis in the spirit of proximal partial identification Ghassami et al. (2023) that here follows the plausibly-exogenous relaxation of an exclusion restriction Conley et al. (2012): posit a direct policy effect on the proxies and carry it through to the estimand. Suppose it fails: general-equilibrium spillovers from the closure depress or inflate the insulated sectors, so that over the post-period $p_{jt} = \boldsymbol{\mu}_j^\top \boldsymbol{\lambda}_t + \delta_j d_t + e_{jt}$, with δ_j the direct policy effect on proxy j and $\boldsymbol{\delta} = (\delta_j)_{j\in\mathcal{N}_0}$. Because the bridge weights are estimated over the pre-period, where $d_t = 0$ and the contamination is absent, $\widehat{\mathbf{w}}$ is untouched and the violation reaches the estimate only through the post-period counterfactual, as the

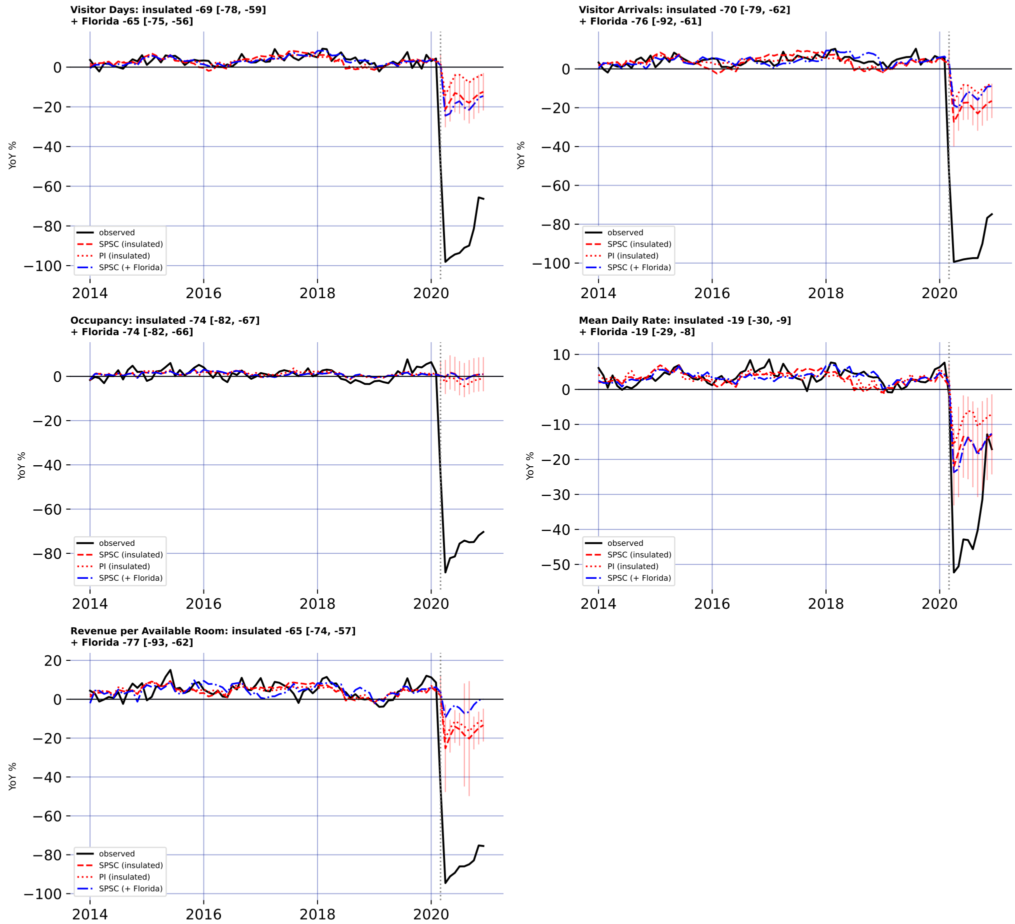

Figure 3: Observed, SPSC and PI insulated counterfactuals, and SPSC + Florida, for five outcomes.

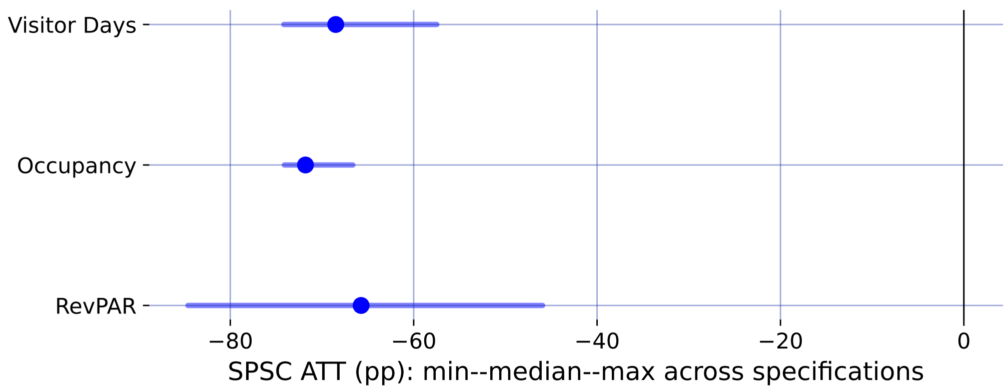

Figure 4: Specification robustness band: median SPSC ATT (point) and min–max range across eight specifications.

exact bias

$$\widehat{\text{ATT}} = \text{ATT} - \boldsymbol{\delta}^\top \widehat{\mathbf{w}}, \tag{5.9}$$

so the bias is linear in the contamination and known up to $\boldsymbol{\delta}$. This makes the sensitivity fully transparent.

For Visitor Days the fitted weights sum to 1.92, so a common spillover of δ points moves the ATT by $1.92\,\delta$. Driving the insulated-only estimate of -69 points to zero would take $\delta \approx +36$ percentage points, the wrong sign (a border closure that *raised* health-care, finance, and government employment growth by nearly forty points) and larger than any insulated sector's *entire* observed 2020 movement, which reached at most 12 points and is itself mostly the macro recession the proxies are meant to carry, not policy spillover. In the economically natural direction, the policy depressing the insulated sectors, $\delta_j < 0$, the bias runs the other way: charging each sector's *full* 2020 decline to the border policy moves the estimate to -83 points, so the reported effect would if anything *understate* the truth. Even an adversarial assignment that signs every δ_j to mask the effect, bounded by each sector's observed movement, shifts the ATT by at most 18 points. Occupancy is more robust still: its weights sum to essentially zero (+0.03), so a common exclusion violation cancels outright. The conclusion does not require the insulated sectors to be perfectly insulated, it survives any exclusion violation of plausible magnitude, and the one plausible direction makes it conservative.

Equation (5.9) treats the insulated sectors symmetrically, but one is more suspect than the rest: construction tracks hotel building, so it is the insulated proxy most plausibly linked to tourism and the one a reader is likeliest to distrust. Dropping it outright is the direct negative-control check. Construction is not idle in the bridge, carrying a weight of -0.35 on Visitor Days, yet removing it leaves the estimate at -73 points against -69 with all five sectors, and the pre-treatment fit is if anything slightly better (RMSE 2.52 versus 2.58), because the four remaining sectors span the same macro-recession factor and absorb its removal. The direction is the telling part: the estimate moves *away* from zero, not toward it. Were construction contaminated by the tourism collapse, its co-movement with the treated series would pull the counterfactual down and bias the effect toward zero, so purging it would enlarge the effect, which is exactly what happens. The headline does not lean on the one donor a skeptic would challenge first.

5.4.3. A placebo in time A distinct worry concerns the estimator itself: does the ridge-GMM bridge, with its spline de-trending, manufacture a large effect out of ordinary pre-period variation? The standard falsification is an in-time placebo, assign a fake border closure in a pre-pandemic March, truncate the sample before 2020 so no real shock is in view, and refit. A design that recovers effects only where they exist should return roughly zero.

It does. Across Visitor Days, Occupancy, and Revenue per Available Room, and fake treatment dates in March of 2017, 2018, and 2019, the placebo ATTs never exceed 4 points in magnitude, the scale of the pre-treatment RMSE, and an order of magnitude below the ~ 70-point effects the same procedure returns for 2020. The bridge does not fabricate a collapse out of business-as-usual fluctuation, so the 2020 estimate is a property of the treatment, not of the estimator.

5.5. *Who bore the cost: incidence across islands and hotel tiers*

The DBEDT workbook that supplies the statewide hotel outcomes also reports them by county and, for revenue per available room, by hotel class, so the same proximal machinery can ask not only how large the tourism collapse was but who absorbed it. For each disaggregated series I refit SPSC on the identical insulated donors and, as in Section~5.3, on the insulated set plus Florida; Figure 5 collects the estimates. Three readings stand out.

First, the demand shock was close to spatially uniform. Hotel occupancy fell between 65 and 76 points across all four counties, so the intuition that the tourism-dependent neighbor islands suffered a deeper demand hit than Oahu is not borne out in occupancy: the rooms emptied everywhere at once.

Second, the collapse ran through volume, not price. The average daily room rate fell only 12 to 24 points, a fraction of the occupancy drop, so hotels largely held their posted rates while rooms went unsold. Because revenue per available room is occupancy times rate, the revenue loss is almost entirely the occupancy loss.

Third, the revenue burden fell regressively across the market. RevPAR by hotel class traces a clear gradient: the luxury and upper-upscale tiers lost the most (90 and 99 points), the upper-midscale tier markedly less among the well-fit segments (56), with midscale and economy least affected of all. Discretionary luxury-leisure travel evaporated where lower-tier and essential lodging demand partly persisted, so the establishments and workers tied to the top of the market bore the sharpest revenue hit.

These disaggregated panels are noisier than the statewide aggregates, and several cells (faded in Figure 5, pre-treatment RMSE above six growth-rate points) carry the near-zero GMM standard errors that Section~3.2 flags as fragile when the proxies supply one instrument against the donors. I read those cells as directional only; the three patterns above rest on the well-fit segments and hold under both the insulated bound and the Florida-adjusted point.

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6. DISCUSSION

The methodological lesson of Hawaii is that a good pre-treatment fit is not identification when a policy and a global shock arrive in the same month. A synthetic control that matches Hawaii to any pre-pandemic counterfactual, whether by borrowing across states or, as own-history extrapolation does, across the treated unit's own past, produces a clean-looking counterfactual and a large ATT, but that ATT is the border-closure effect plus the pandemic's contribution to Hawaii's outcomes, and nothing in the fit reveals the mix. Proximal / negative-control inference is what separates them: instrumenting the latent pandemic factor with insulated employment sectors that carry it but are immune to the policy, both headline specifications, SPSC and base PI, net out the macro-recession component and agree on the result, so the estimate rests on the identification rather than on any single estimator.

At the same time, the analysis is candid about the limit of what the shipped data can identify. The insulated sectors span the pandemic's general-recession factor but not its

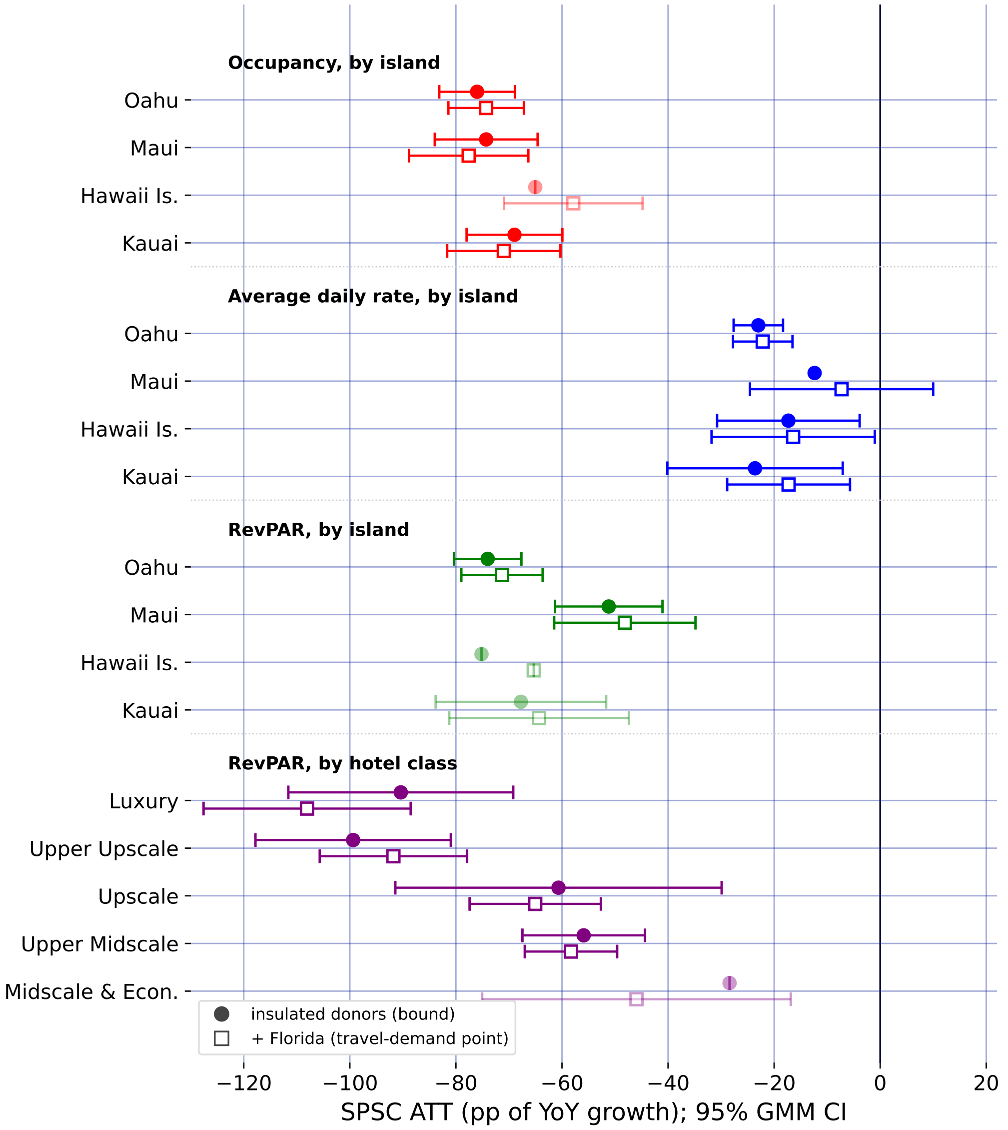

Figure 5: Subgroup ATTs by island and hotel class: insulated bound (circles), Florida-adjusted point (squares).

travel-demand-collapse factor: even with an open border, Hawaii tourism would have fallen sharply as people stopped flying. SPSC on insulated sectors alone therefore recovers only the macro component of the counterfactual, and the resulting tourism ATTs over-attribute the collapse to the policy; they are upper bounds in magnitude, which is why I report them as a band rather than as points. Recovering an actual point estimate from insulated donors alone is not possible here, because every treated series loads on the travel-demand factor those donors miss; the point has to be spanned from outside the treated economy, which is what the open-destination donors do. The substantive conclusion is thus deliberately modest and, I would argue, more credible for it: the border closure coincided with a near-total collapse in measured tourism demand, of which a bounded share is causally the policy, with a convergent ensemble of external travel-demand donors pulling the Visitor Days estimate off its upper bound toward a policy-centered band.

Section~5.3 takes the decisive step of closing the two-factor gap with external travel-demand negative controls (inbound air travel to Florida and to a set of other exposed-but-open leisure markets that did *not* lock down), and the result is instructive about both the promise and the care the approach demands. For Visitor Days the open-destination donors do what the design promises, and they do it *in agreement*: fit one at a time, all four independently pull the estimate off the insulated-only bound into a policy-centered band, each improving the pre-treatment fit. Because the donors are chosen independently (different cities, two different states), their convergence is far stronger evidence that the missing factor is real and separable than any single proxy could be. A well-chosen ensemble also reaches further than one volume series: Phoenix, whose resort room rates track leisure demand, pulls the daily-room-rate series that Florida (a volume proxy) leaves essentially untouched. But the ensemble does not mechanically dissolve every bound, and the search must stay disciplined: near-collinear open destinations added *en masse* induce large offsetting weights and hurt the fit, so the donors are best read as a corroborating convergence band; and Puerto Rico, tempting as a fellow island, fails as a control because it imposed its own stay-at-home order in mid-March 2020 and is a contaminated, treated-like unit that, if anything, bounds the effect from the other side.

These estimates are one side of a ledger. They price the short-run economic cost of the border closure through the end of 2020; they do not, on their own, value the public-health benefit (lives saved, hospital strain averted) the closure was meant to buy. A bounded cost, though, is exactly the input a cost-benefit calculation needs, and the surrounding literature supplies the other side. Work on early containment finds that acting even days sooner averts a large share of deaths: Bayat et al. (2020) estimate that locking down ten days earlier could have cut New York deaths by more than 80%. Evidence on comparably aggressive closures finds the economic damage sharp but short-lived Ke and Hsiao (2022). Set side by side, the calculus for a decisive closure is a large but transitory economic loss weighed against a benefit measured in lives, and this paper supplies the first credible, bounded estimate of that loss for the U.S. case. The incidence analysis of Section~5.5 adds that the loss fell unevenly: hardest on the upper tiers of the lodging market, and through lost volume rather than lower prices. What the paper also provides is a template for pricing such policies at all: when the intervention and the crisis it responds to are one and the same event in time, no comparison unit is clean, and the analyst's task is to find a variable that sees the crisis but not the policy.

7. CONCLUSION

This paper estimates the short-run economic consequences of Hawaii's March 2020 mandatory quarantine while confronting the identification problem that its predecessors on this topic could not: for a single treated unit, the border-closure policy and the COVID-19 pandemic are perfectly collinear in time, so no synthetic control that merely matches Hawaii to a pre-pandemic counterfactual can separate the two. I resolve this with proximal inference (negative-control estimators) in two specifications (Single Proxy Synthetic Control and base Proximal Inference) that instrument the latent pandemic factor using insulated employment sectors which carry the pandemic's macro-recession shock but are immune to the border policy.

Estimated by both specifications, the insulated-sector proxies identify a large, robust tourism-demand collapse (about 60–75 points of year-over-year growth for the visitor-volume series). But because those sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly: the insulated-only estimates are upper bounds in magnitude on the policy effect, reported as a robustness band rather than as points. Because every treated series loads on the travel-demand factor those donors miss, the trustworthy point comes only once the missing factor is spanned from outside the treated economy, the role of the external open-destination travel-demand donors.

Methodologically, the study demonstrates how negative-control inference yields credible causal estimates when a policy and a global shock coincide, a structure common to pandemic border closures, but also to climate shocks and other systemic crises where a would-be counterfactual unit is itself engulfed by the event. Substantively, it delivers a bounded, honest estimate of the economic price of Hawaii's "Great Isolation with Aloha characteristics' '. Adding a small ensemble of external no-lockdown travel-demand donors (inbound air travel to Florida and to the Tampa, Orlando, and Phoenix leisure markets) takes the step from a bound to a policy-centered band: fit independently, they converge in pulling the Visitor Days effect toward the policy-only magnitude, and one of them (Phoenix) reaches the room-rate series a single volume proxy cannot. Finishing the job for every outcome asks only for more such open-destination proxies, disciplined by relevance, which the design readily accommodates.

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